

U.S. Multi Industry & Electrical Equipment

US Multis: Mapping the Data Center Service Footprint for Chiller OEMs (CARR / JCI / TT)



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A common question we hear from investors is which HVAC OEM possesses the strongest services infrastructure for supporting the rapidly growing North American data center market. In this note, we evaluate the HVAC service footprints of Trane, Johnson Controls, and Carrier across North America by mapping their commercial service locations against the Aterio North America Data Center database. Using latitude and longitude coordinates for both service branches and active data center sites, we assess proximity coverage across ~2,163 operational data centers representing ~52GW of installed capacity across US and Canada (this does not include announced but non-operational projects). While service quality remains a frequent topic of debate amongst investors and customers, our analysis focuses solely on geographic accessibility and coverage.

JCI leads on footprint density, but all three OEMs appear well positioned to service existing data center capacity: From a footprint perspective, Johnson Controls stands out with the largest commercial service network in our analysis at 270 locations, nearly double Trane's 143 and more than triple Carrier's 88. This scale advantage translates into superior immediate proximity to data centers (with JCI owning the largest share of "closest" service locations both on a site count and installed capacity basis). **JCI is the nearest provider for ~48% of active sites and ~47% of installed capacity** (versus ~25%/~24% for Trane and ~27%/~29% for Carrier).

However, the practical differences narrow meaningfully once we expand the service radius beyond the nearest location. While JCI achieves the strongest overall coverage profile, all three companies cover the vast majority of North American data centers. Within a 30-mile radius, Trane covers ~81% of active sites, JCI ~84%, and Carrier ~76%. When weighted by installed capacity, all three companies maintain access to ~70%+ of active megawatt capacity. Median distance to an active data center is ~10.7 miles for JCI versus ~14.7 miles for Trane and ~14.4 miles for Carrier, reinforcing JCI's density advantage while also highlighting that all three companies' service locations are generally clustered close to existing data center deployments.

Geography remains concentrated, but whitespace opportunities are still available: North American data center capacity remains highly concentrated geographically, with ~75% of installed capacity located across just ten states. Virginia, Texas, Oregon, and Ohio alone account for ~50% of installed capacity. Not surprisingly, service coverage across these major markets is generally strong, although leadership varies by state and metric. A key takeaway is also on what remains uncovered: **~10GW of active capacity, (~20% of total installed North American capacity), sits outside a 30-mile radius of any Trane, JCI, or Carrier commercial service location.** This equates to roughly ~10% of active sites and represents a meaningful whitespace opportunity for all three companies. As data center construction expands beyond traditional hubs and into emerging geographies, we believe that **service capacity additions, branch expansions and technician deployment in these underserved markets could become an increasingly important source of competitive differentiation going forward.**

BERNSTEIN TICKER TABLE

Ticker	Rating	Cur	14 Aug 2026	Price Target	TTM	Adjusted EPS				Adjusted P/E (x)		
			Closing Price		Rel. Perf.	Cur	2025A	2026E	2027E	2025A	2026E	2027E
TT (Trane)	O	USD	480.20	575.00	(7.0)%	USD	13.06	15.28	17.79	36.8	31.4	27.0
JCI (Johnson Controls)	O	USD	153.64	190.00	28.1%	USD	3.78	5.09	6.35	40.7	30.2	24.2
CARR (Carrier)	M	USD	62.78	78.00	(24.4)%	USD	2.57	2.85	3.29	24.4	22.0	19.1
SPX			7,785.76									

O - Outperform, M - Market-Perform, U - Underperform, NR - Not Rated, CS - Coverage Suspended

Source: Bloomberg, Bernstein estimates and analysis.

INVESTMENT IMPLICATIONS

We rate TT (PT \$575) and JCI (PT \$190) Outperform.

We rate CARR (PT \$78) Market-Perform.

DETAILS

We note that service organizations across the HVAC industry operate differently across companies, and technicians are generally (but not always) dispatched directly from commercial service branches. In some cases, OEMs also have dedicated DC service teams over located outside existing sites (e.g., we have heard this for Carrier). In this analysis, we focus exclusively on commercial service locations (identified from individual company websites) and exclude independent dealers, distributors, franchise partners, and parts stores.

METHODOLOGY

We collected commercial HVAC service locations for Trane, Johnson Controls, and Carrier directly from company-operated service location branch details available on each company's website. Locations identified as commercial service branches were then geocoded into their respective latitude and longitude coordinates. Independent dealers, distributors, franchise operators, and parts stores were excluded from the analysis in order to maintain consistency across the three companies.

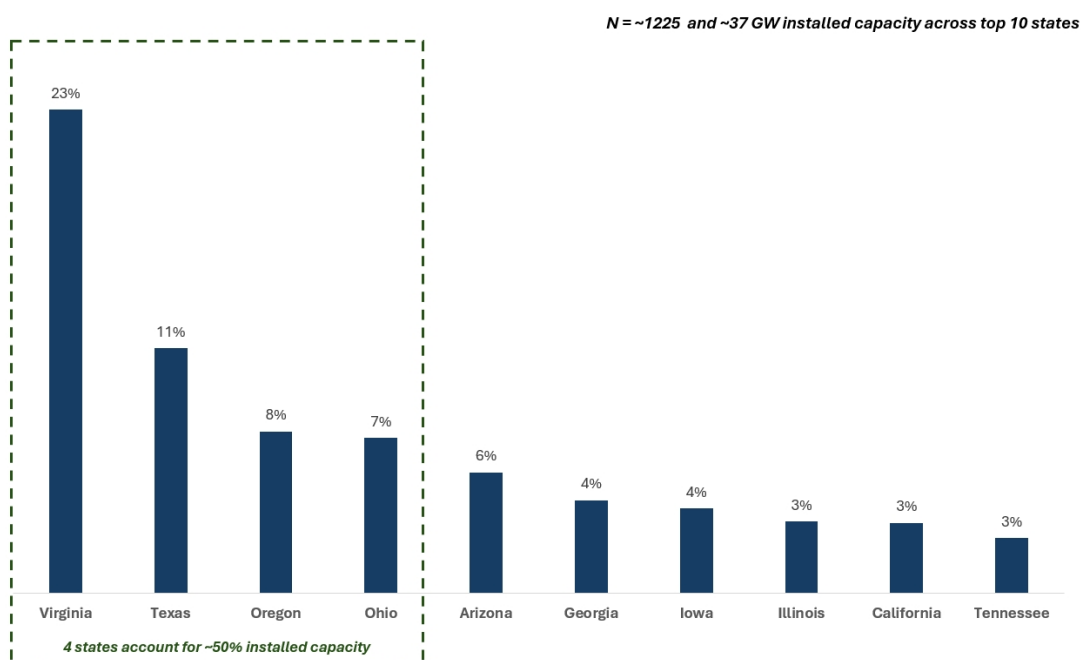
We then accessed the Aterio North America Data Center Dataset (August 2026), focusing exclusively on active operational data center facilities across North America (United States and Canada). The resulting dataset included ~2,163 active data center sites representing ~52GW of active installed capacity across US and Canada.

For each active data center location, we calculated point-to-point distances to all identified Trane, Johnson Controls, and Carrier service branches by mapping the latitude and longitude coordinates for each location. Service coverage was then segmented into different distance ranges basis proximity (<10 miles, 10-30 miles, 30-60 miles, 60-90 miles, 90-120 miles, 120-150 miles, and >150 miles) for the analysis. The analysis was performed on both:

- **Site count basis:** each active data center counted equally
- **Installed capacity basis:** each site weighted according to its active installed megawatt capacity

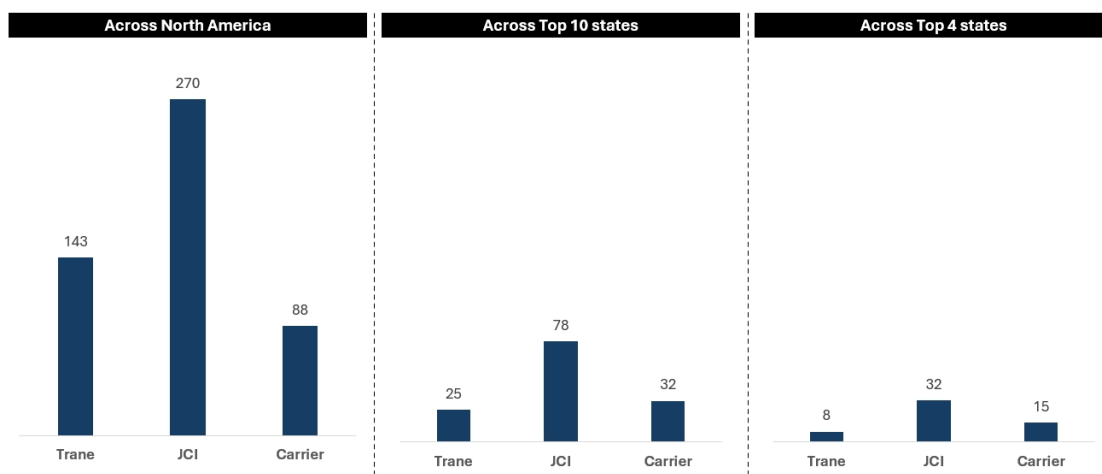
The analysis measures geographic proximity rather than actual service quality, response times, technician availability, customer satisfaction, contractual presence, or dedicated onsite staffing. Investors may thus view these results as a comparative assessment of the access to the physical services footprint (rather than a comprehensive evaluation of service capabilities) for each of the HVAC companies.

EXHIBIT 1: ~75% of active installed data center capacity in North America in 10 states; VA, TX, OR & OH account for ~50%



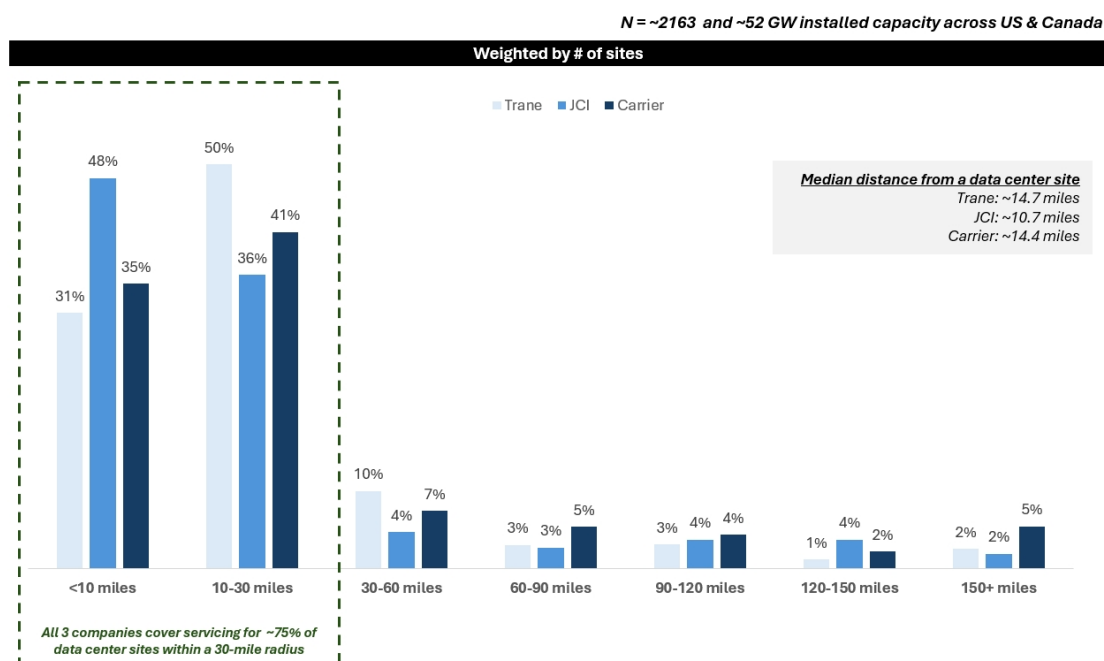
Source: Aterio, Bernstein Analysis

EXHIBIT 2: JCI has the widest services footprint, followed by Trane and then Carrier



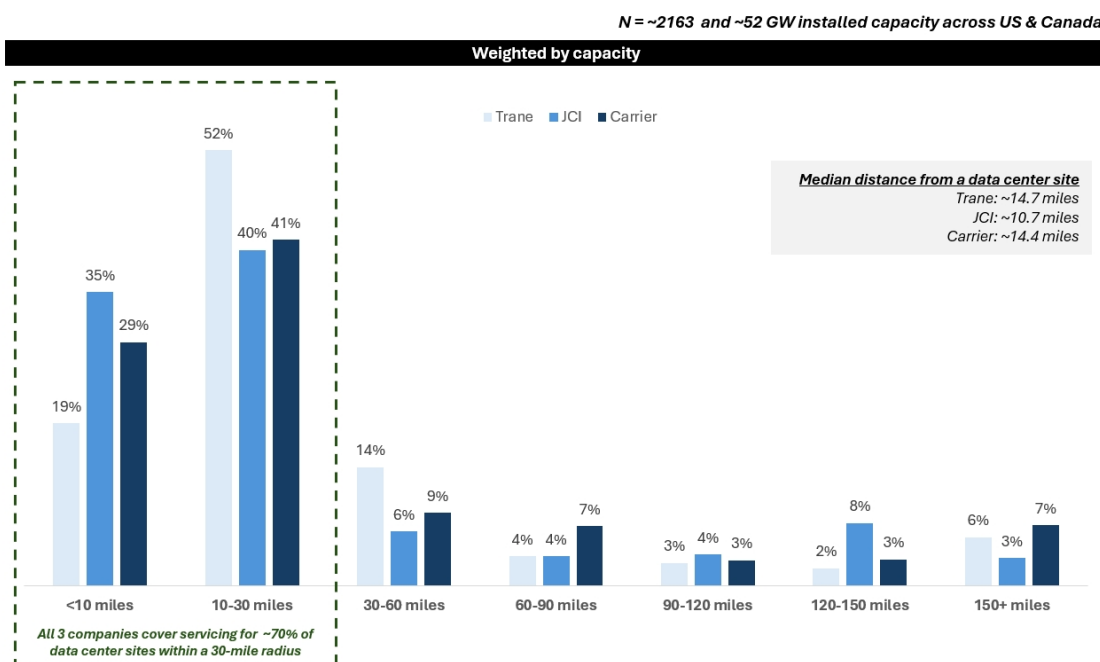
Source: Aterio, Bernstein Analysis

EXHIBIT 3: JCI has the greatest number of service locations in close proximity (<10 miles) to active data centers; but all companies cover existing capacity well (weighted by # of sites)



Source: Aterio, Bernstein Analysis

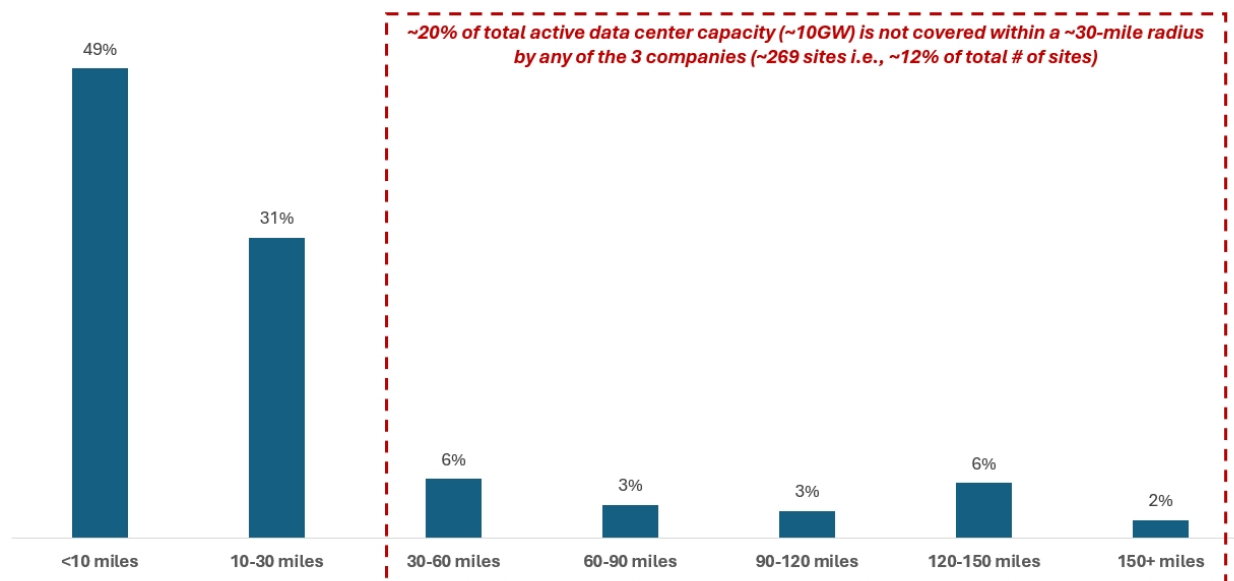
EXHIBIT 4: JCI has the greatest number of service locations in close proximity (<10 miles) to active data centers; but all companies cover existing capacity well (weighted by capacity)



Source: Aterio, Bernstein Analysis

EXHIBIT 5: ~20% of total active data center capacity (~10GW) is a potential whitespace opportunity for Trane, JCI and Carrier

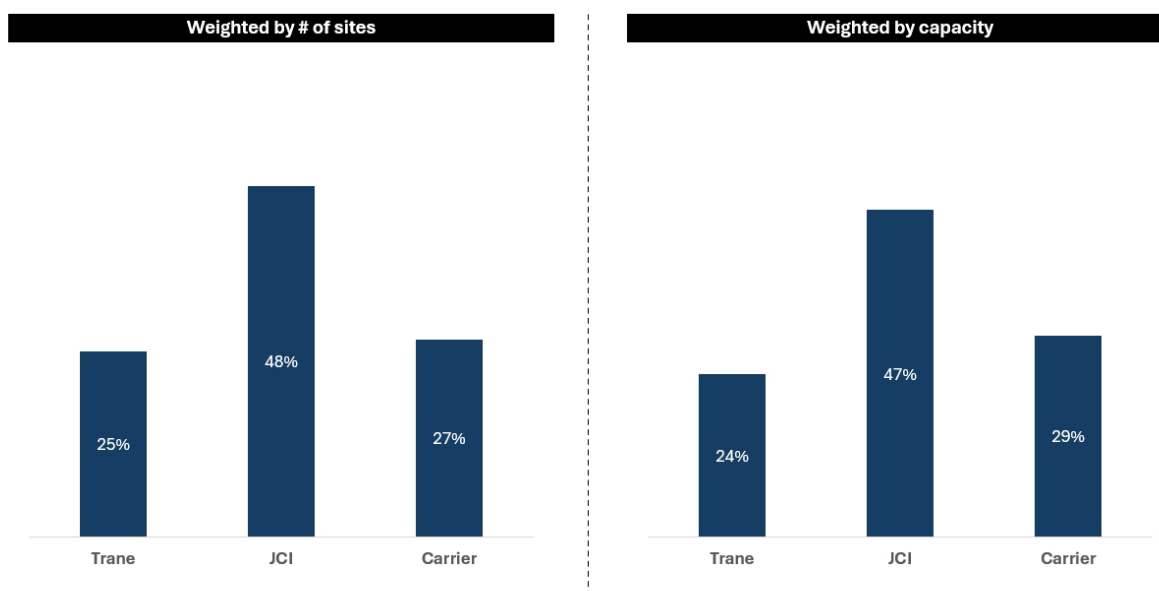
N = ~2163 and ~52 GW installed capacity across US & Canada



Source: Aterio, Bernstein Analysis

EXHIBIT 6: JCI has the “closest” share of service center locations to a data center in North America at ~45%+ both in terms of # of locations mapped and active capacity

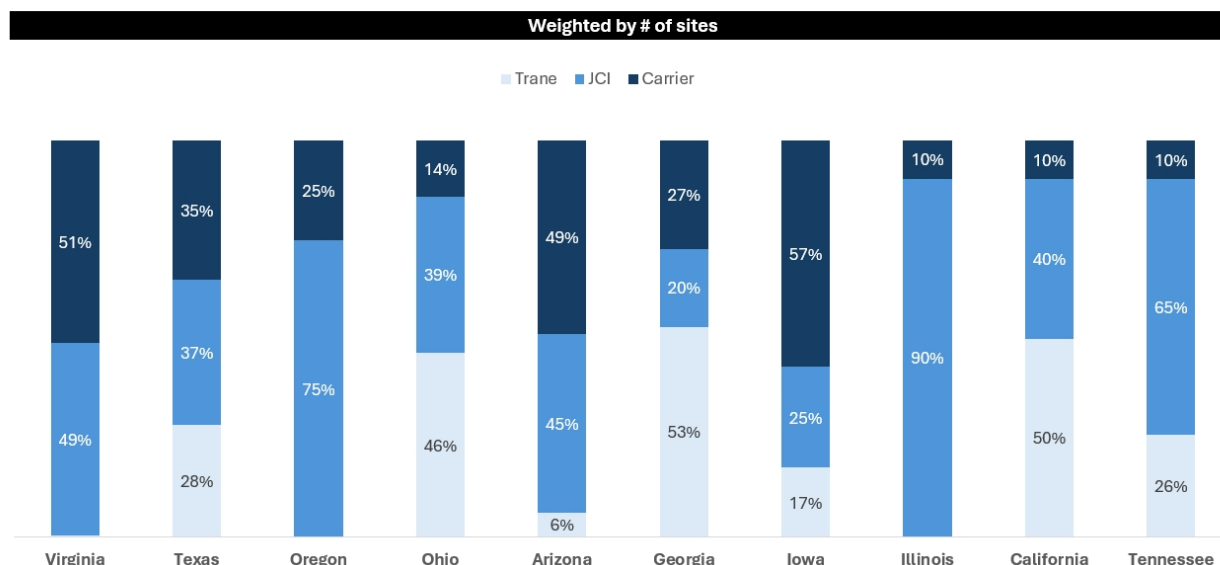
N = ~2163 and ~52 GW installed capacity across US & Canada



Source: Aterio, Bernstein Analysis

EXHIBIT 7: JCI has the “closest” service center locations to an active data center in 4 out of the top 10 states; Carrier and Trane follow with 3 each (when weighted by # of sites)

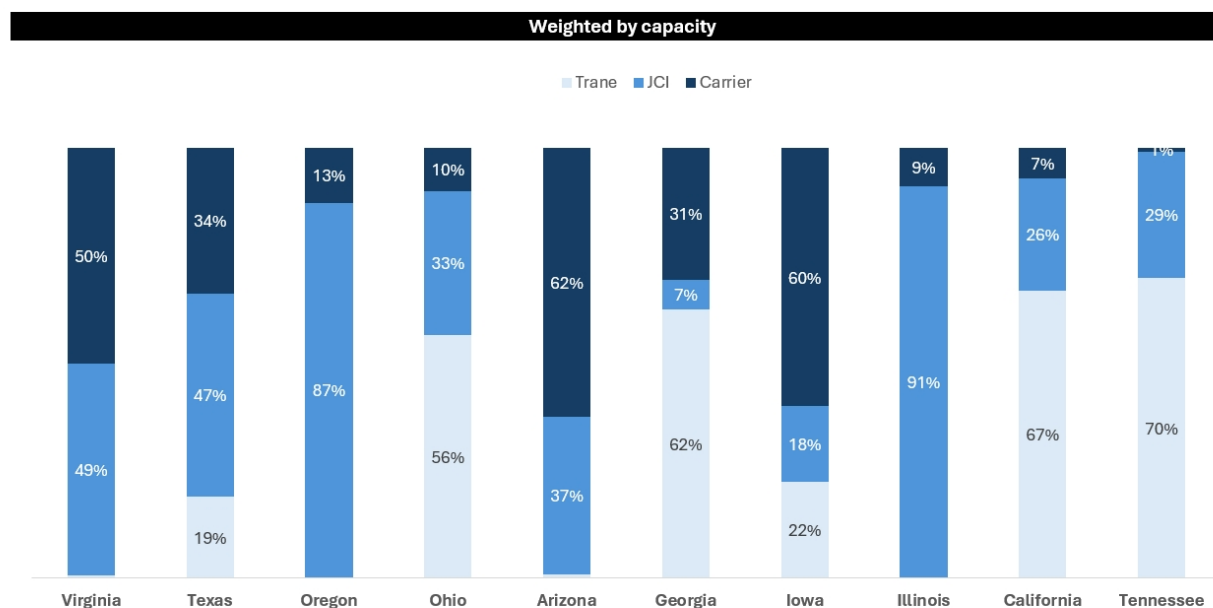
N = ~1225 and ~37 GW installed capacity across top 10 states



Source: Aterio, Bernstein Analysis

EXHIBIT 8: Trane has the “closest” service center locations to an active data center in 4 out of the top 10 states; JCI and Carrier follow with 3 each (when weighted by capacity)

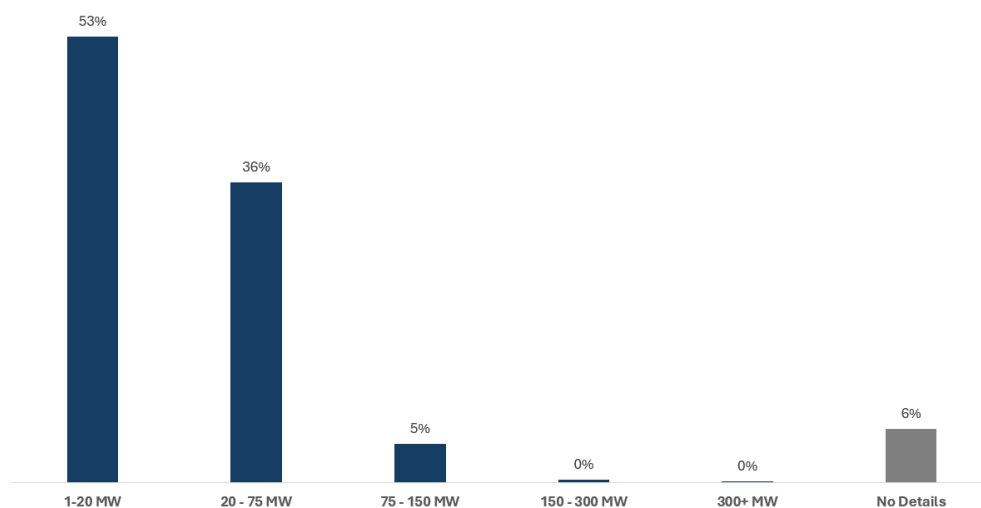
N = ~1225 and ~37 GW installed capacity across top 10 states



Source: Aterio, Bernstein Analysis

EXHIBIT 9: **~90% of active installed data center capacity in North America is <75MW individual builds**

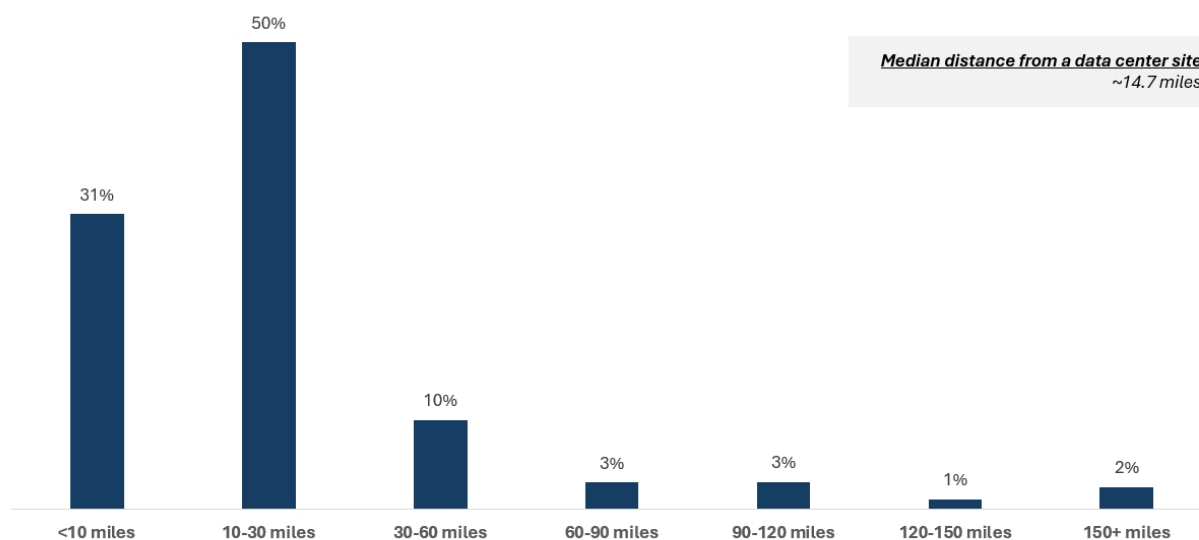
N = ~2163 and ~52 GW installed capacity across US & Canada



Source: Aterio, Bernstein Analysis

EXHIBIT 10: **Trane | ~81% coverage (<30 miles) across North America**

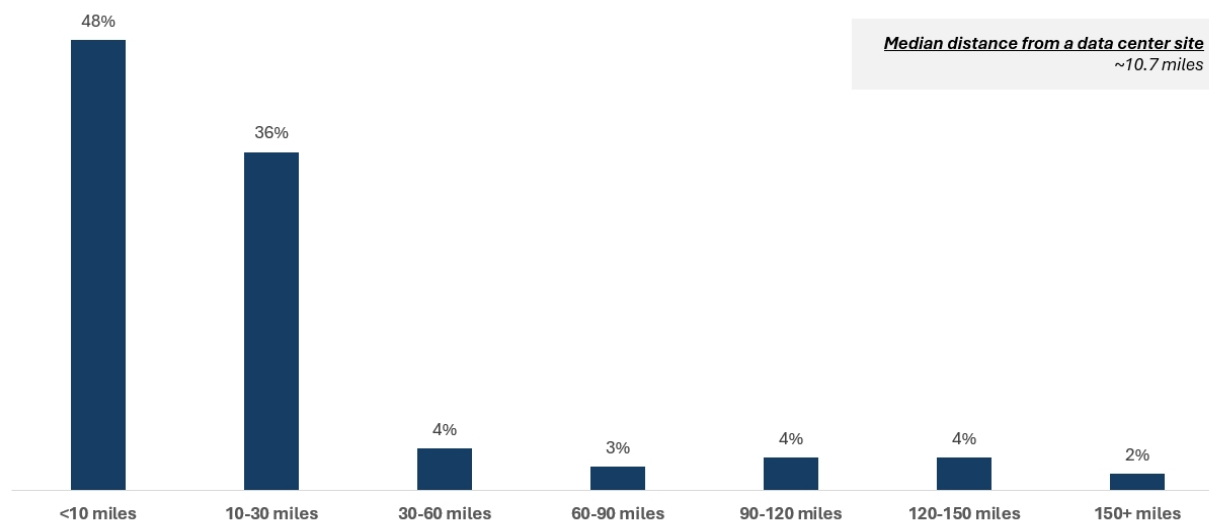
N = ~2163 and ~52 GW installed capacity across US & Canada



Source: Aterio, Bernstein Analysis

EXHIBIT 11: **JCI** | ~84% coverage (<30 miles) across North America

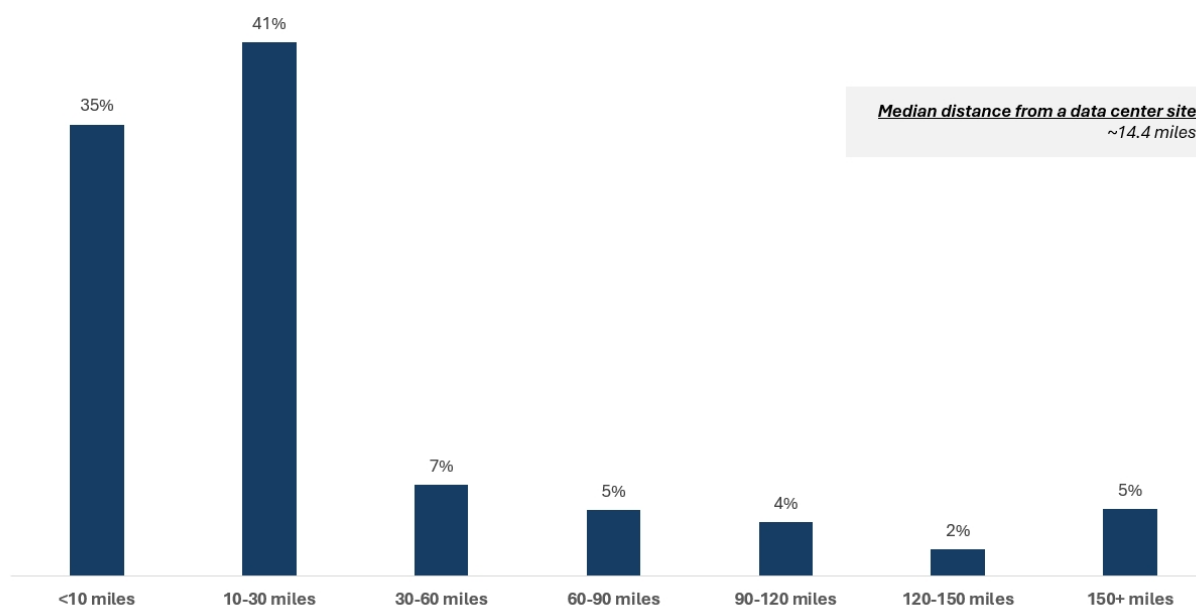
N = ~2163 and ~52 GW installed capacity across US & Canada



Source: Aterio, Bernstein Analysis

EXHIBIT 12: **Carrier** | ~76% coverage (<30 miles) across North America

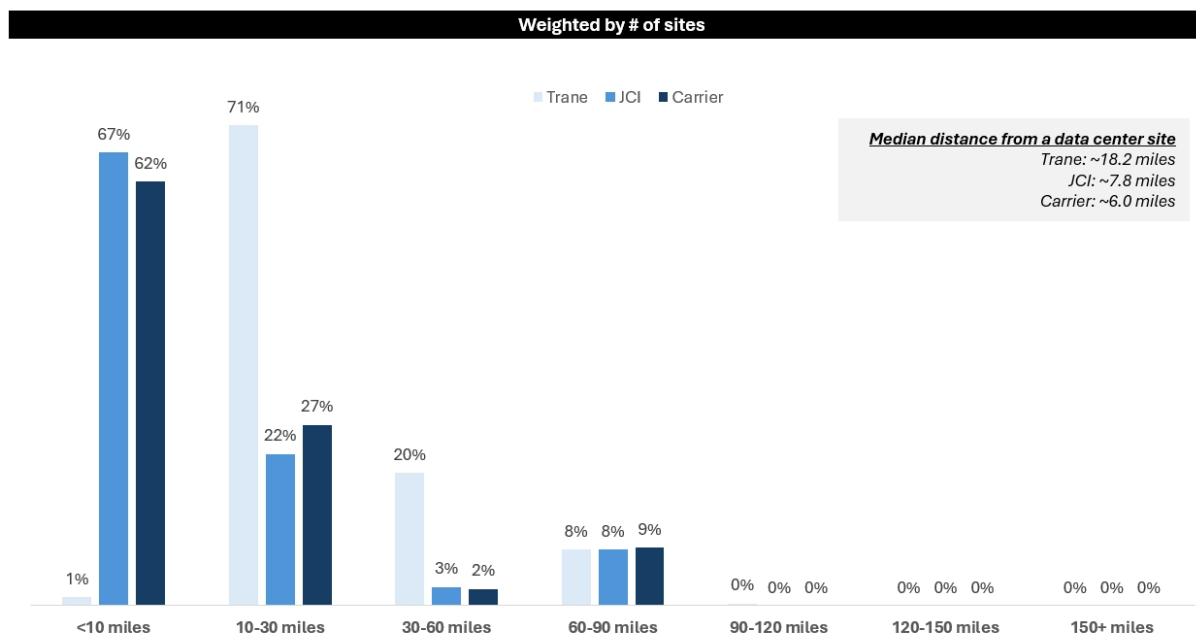
N = ~2163 and ~52 GW installed capacity across US & Canada



Source: Aterio, Bernstein Analysis

EXHIBIT 13: **Virginia | Coverage breakdown weighted by # of sites**

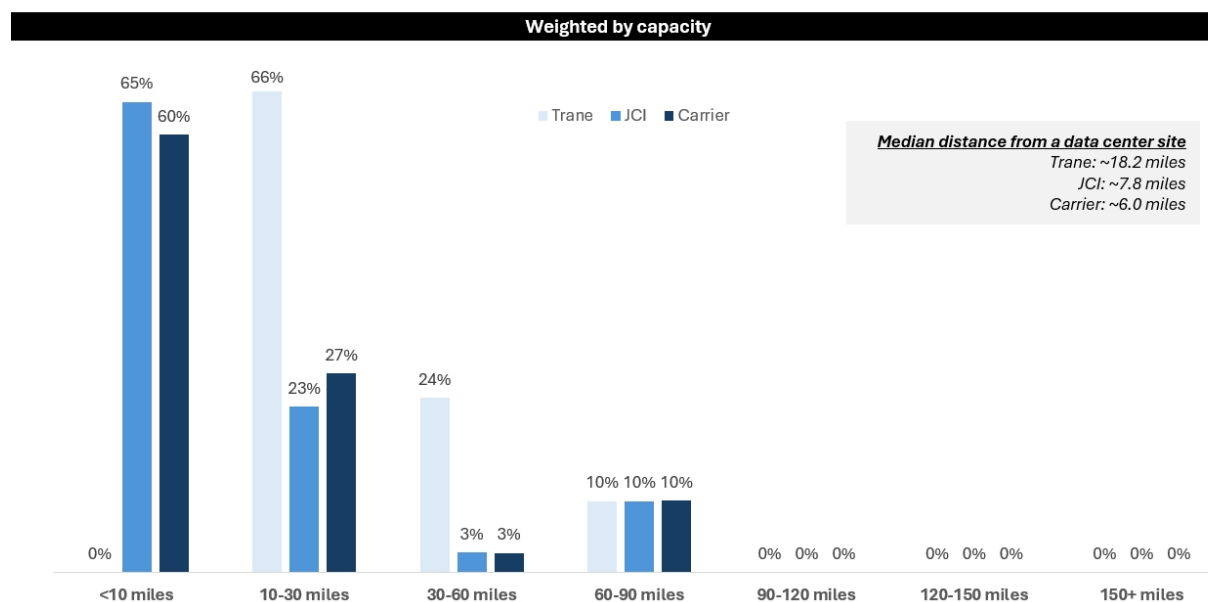
N = ~327 and ~12 GW installed capacity across Virginia



Source: Aterio, Bernstein Analysis

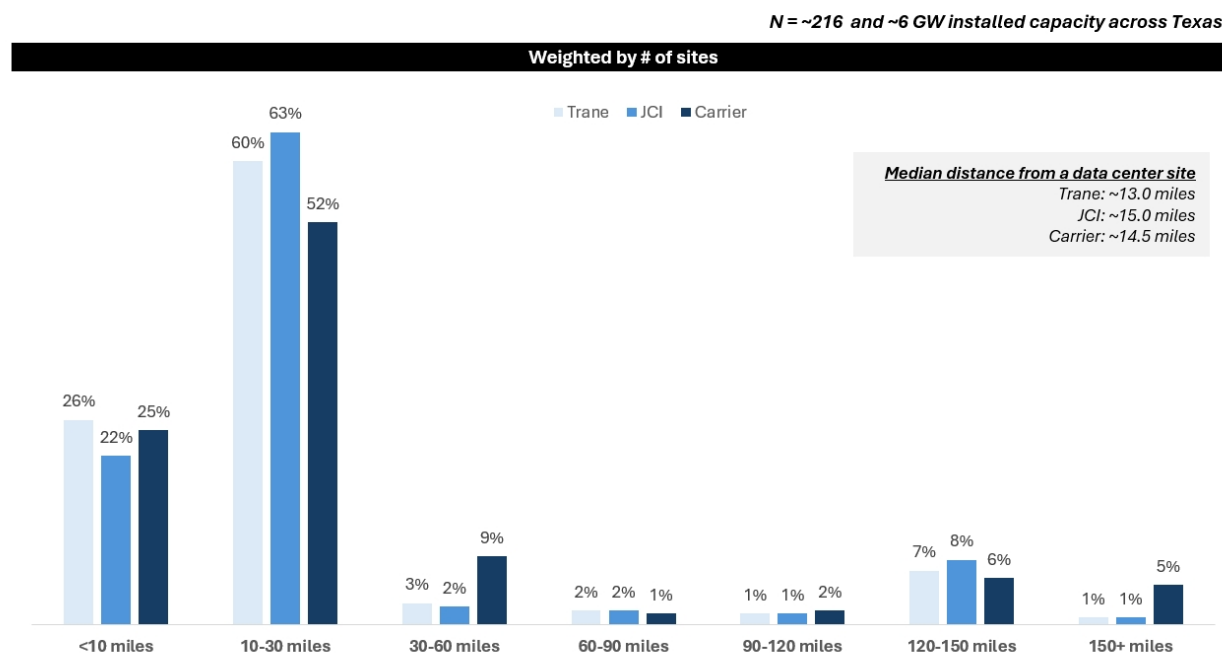
EXHIBIT 14: **Virginia | Coverage breakdown weighted by capacity**

N = ~327 and ~12 GW installed capacity across Virginia



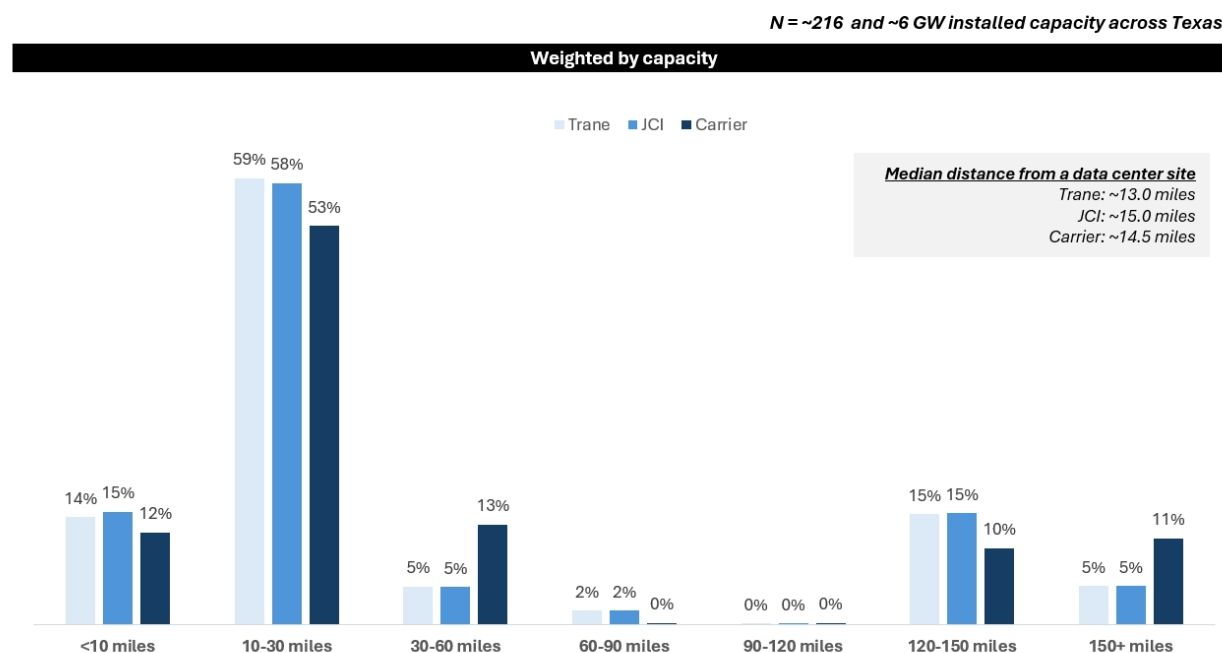
Source: Aterio, Bernstein Analysis

EXHIBIT 15: Texas | Coverage breakdown weighted by # of sites

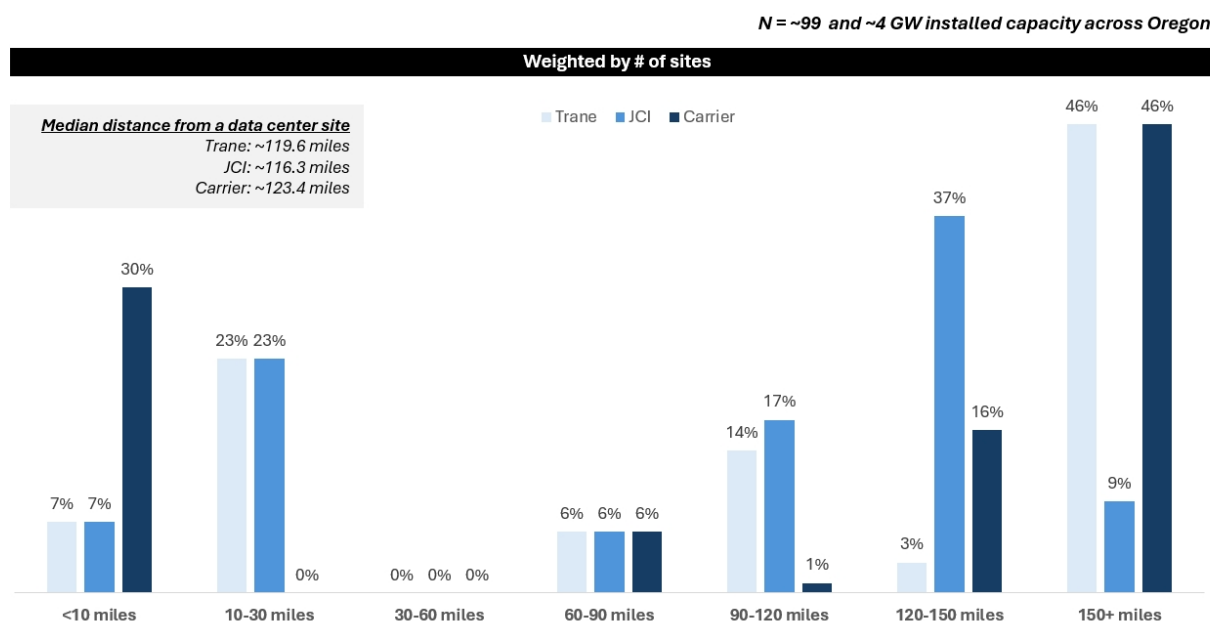


Source: Aterio, Bernstein Analysis

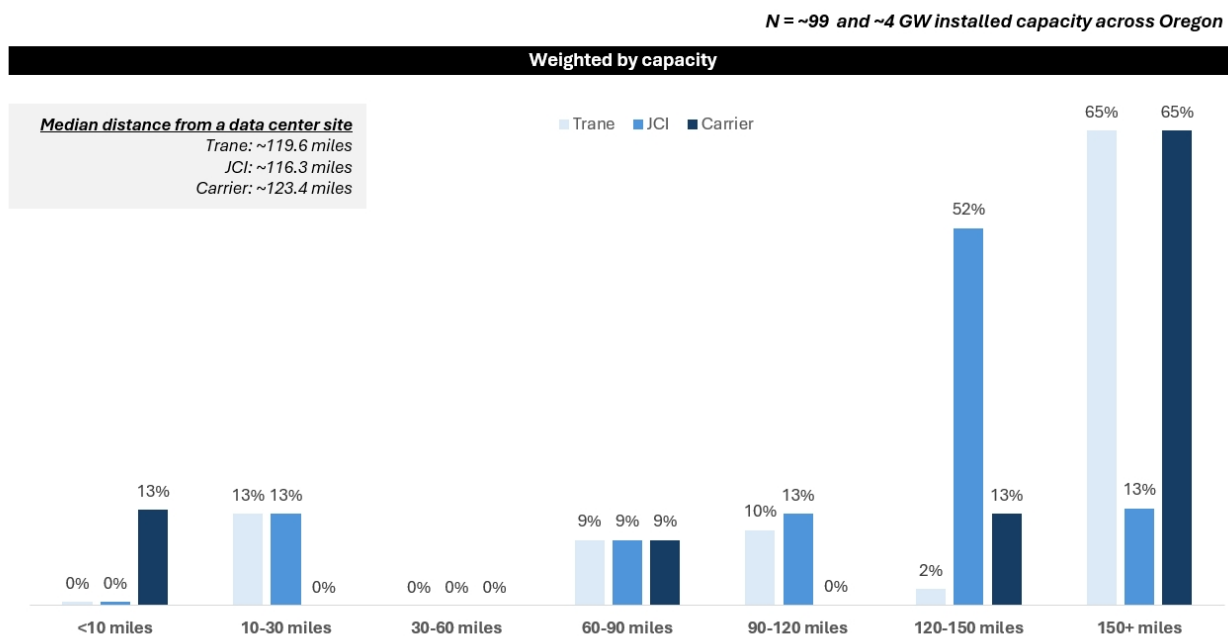
EXHIBIT 16: Texas | Coverage breakdown weighted by capacity



Source: Aterio, Bernstein Analysis

EXHIBIT 17: **Oregon | Coverage breakdown weighted by # of sites**

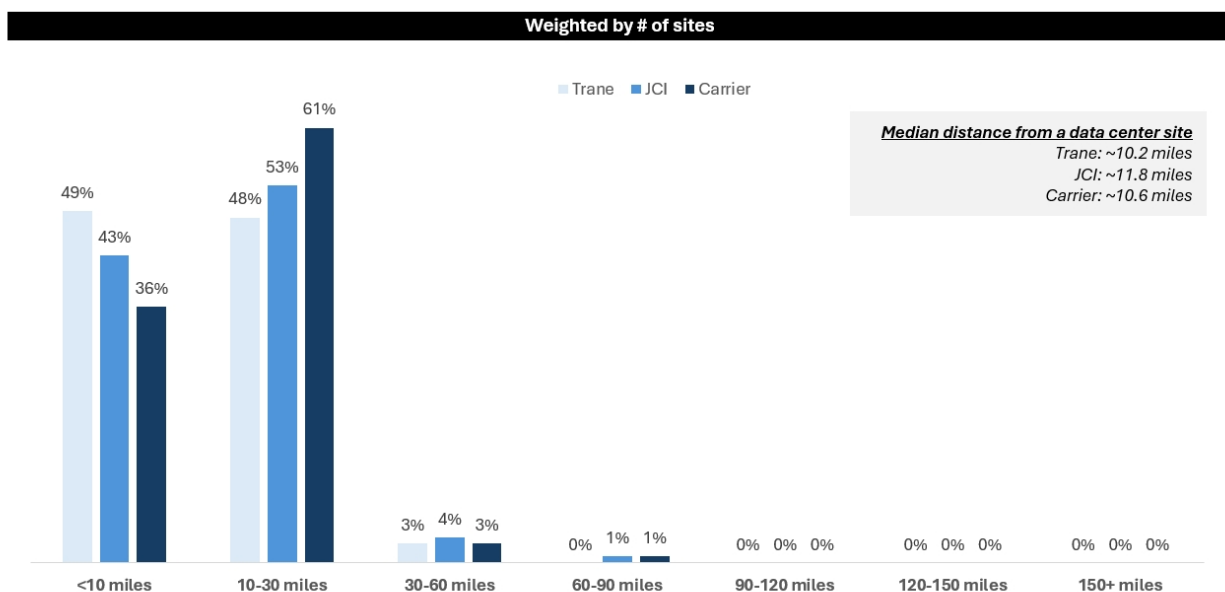
Source: Aterio, Bernstein Analysis

EXHIBIT 18: **Oregon | Coverage breakdown weighted by capacity**

Source: Aterio, Bernstein Analysis

EXHIBIT 19: **Ohio | Coverage breakdown weighted by # of sites**

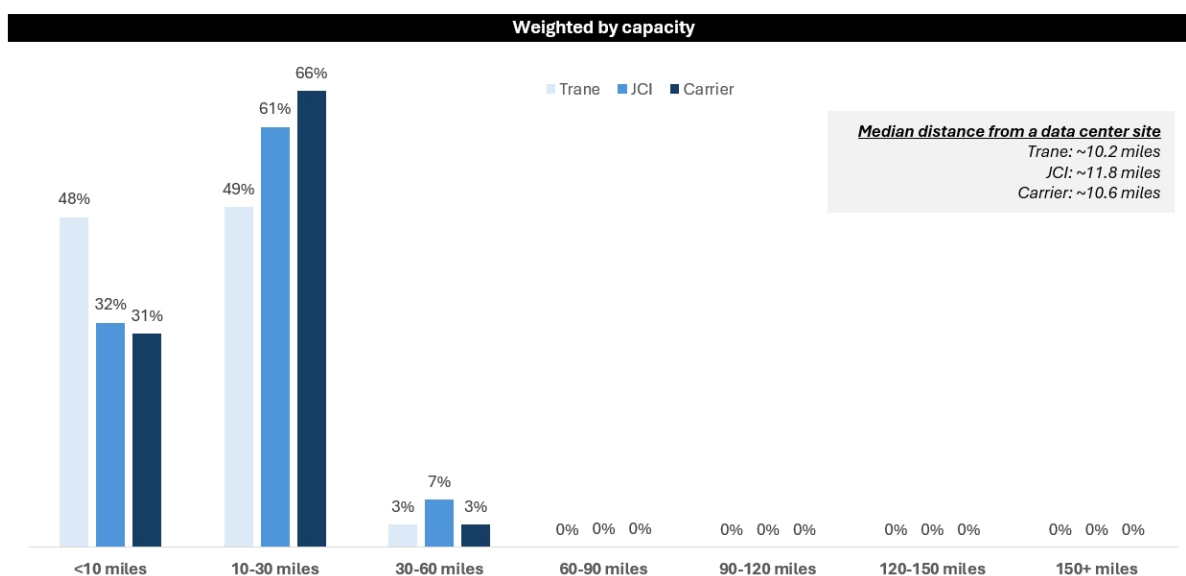
N = ~112 and ~4 GW installed capacity across Ohio



Source: Aterio, Bernstein Analysis

EXHIBIT 20: **Ohio | Coverage breakdown weighted by capacity**

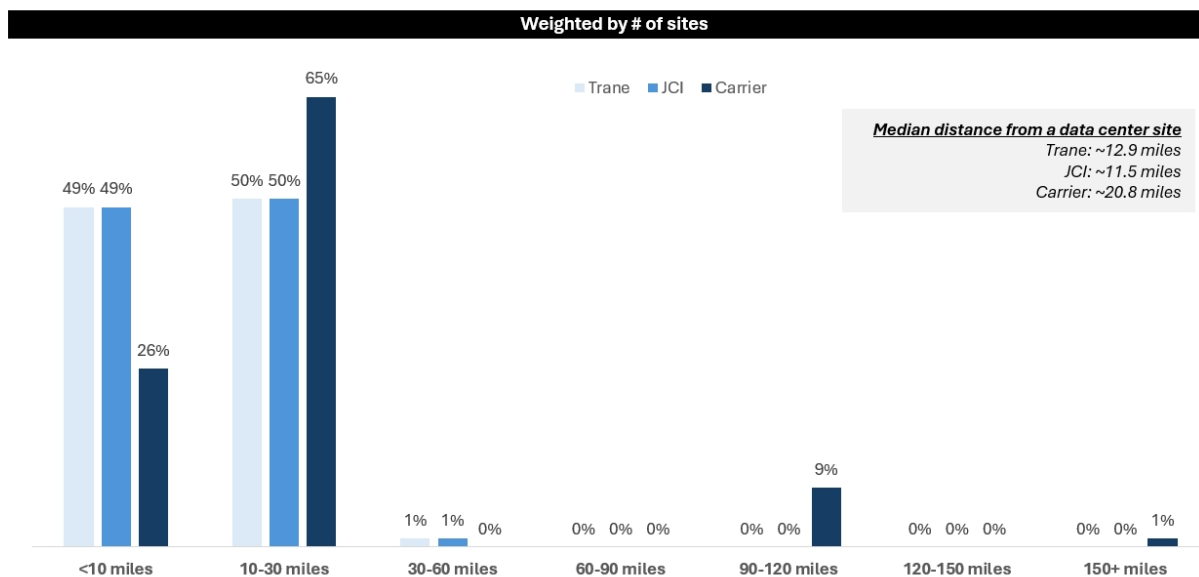
N = ~112 and ~4 GW installed capacity across Ohio



Source: Aterio, Bernstein Analysis

EXHIBIT 21: **Arizona | Coverage breakdown weighted by # of sites**

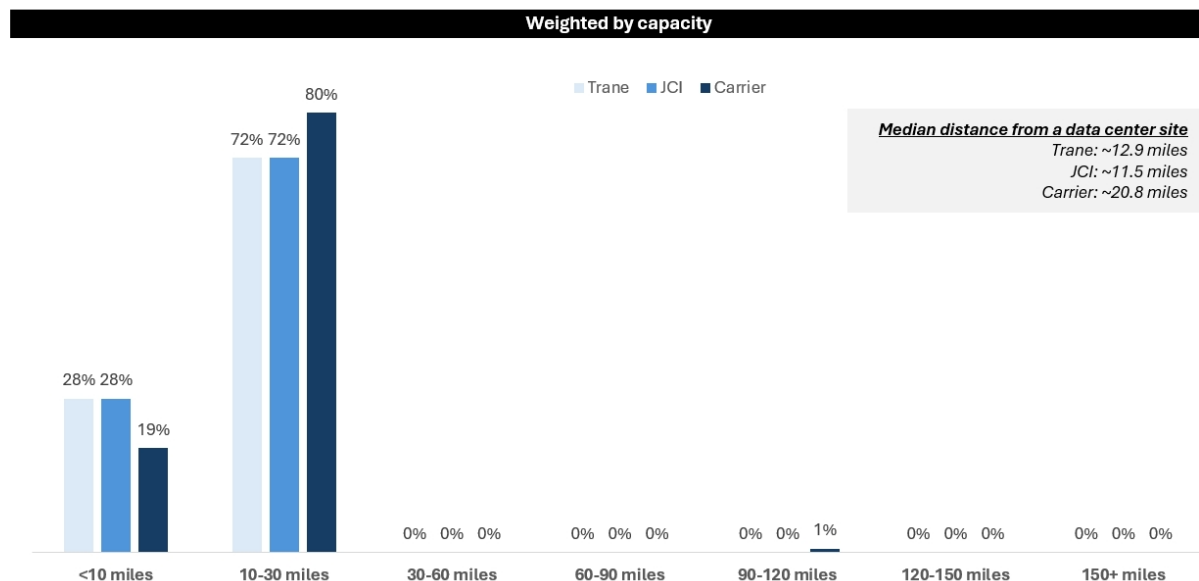
N = ~82 and ~3 GW installed capacity across Arizona



Source: Aterio, Bernstein Analysis

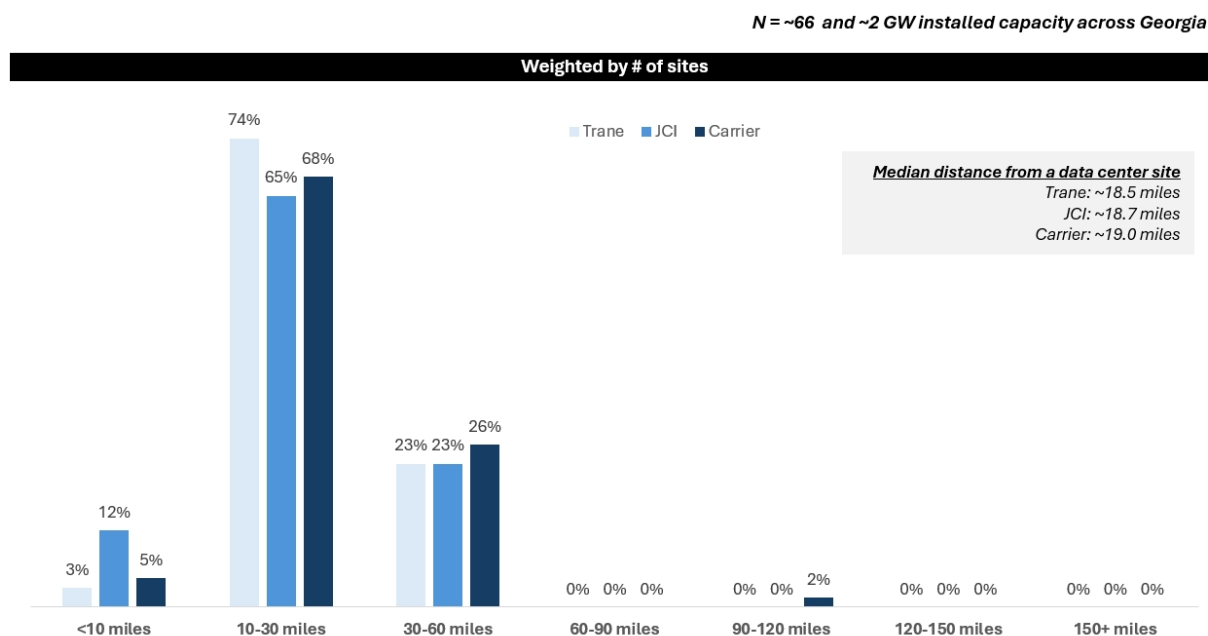
EXHIBIT 22: **Arizona | Coverage breakdown weighted by capacity**

N = ~82 and ~3 GW installed capacity across Arizona



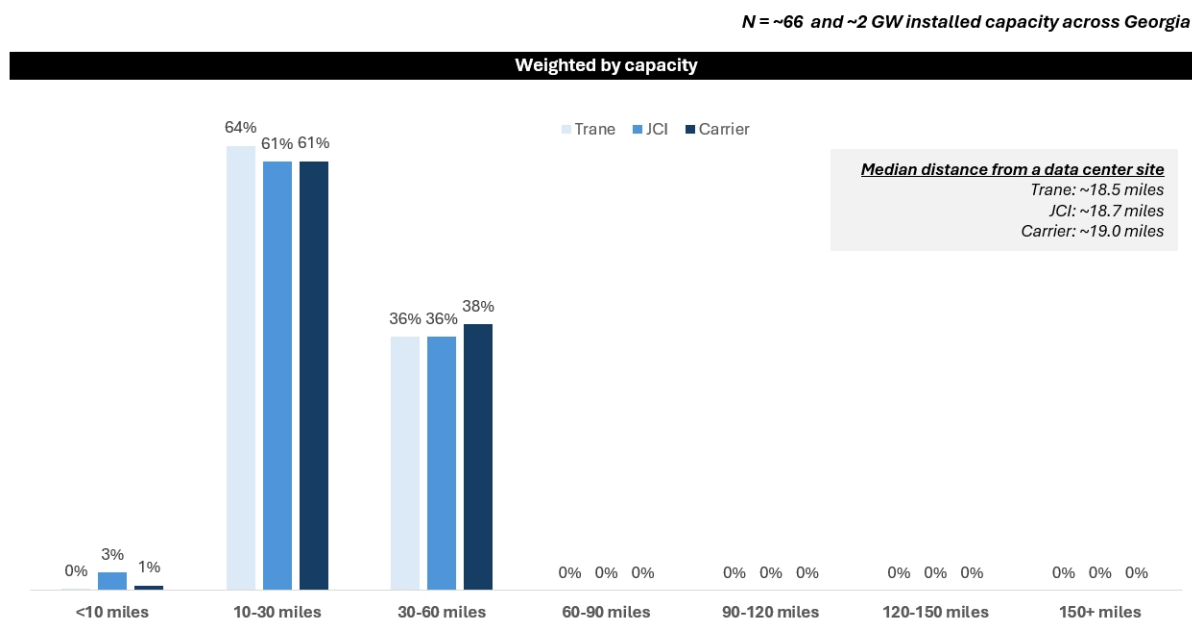
Source: Aterio, Bernstein Analysis

EXHIBIT 23: **Georgia | Coverage breakdown weighted by # of sites**



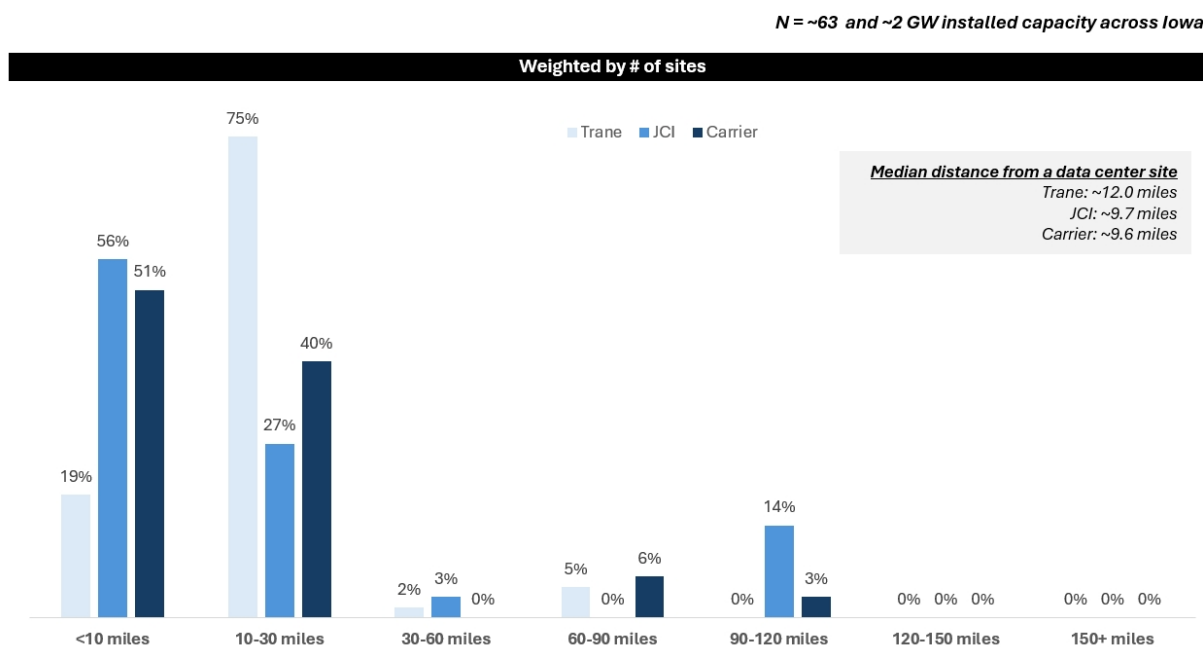
Source: Aterio, Bernstein Analysis

EXHIBIT 24: **Georgia | Coverage breakdown weighted by capacity**



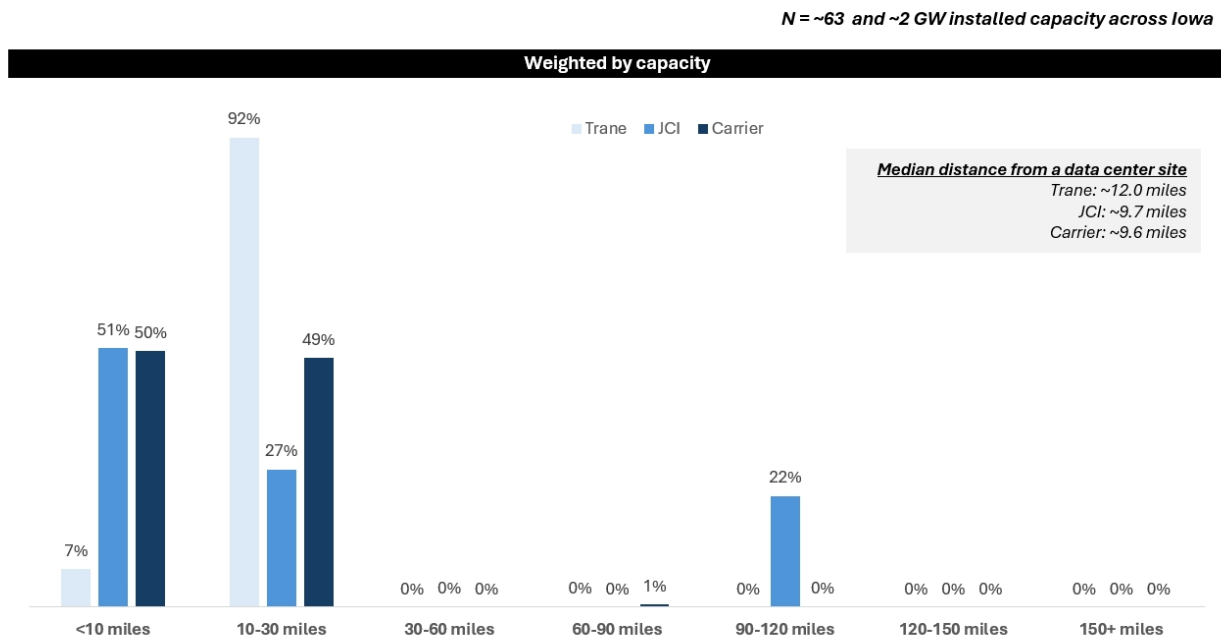
Source: Aterio, Bernstein Analysis

EXHIBIT 25: Iowa | Coverage breakdown weighted by # of sites



Source: Aterio, Bernstein Analysis

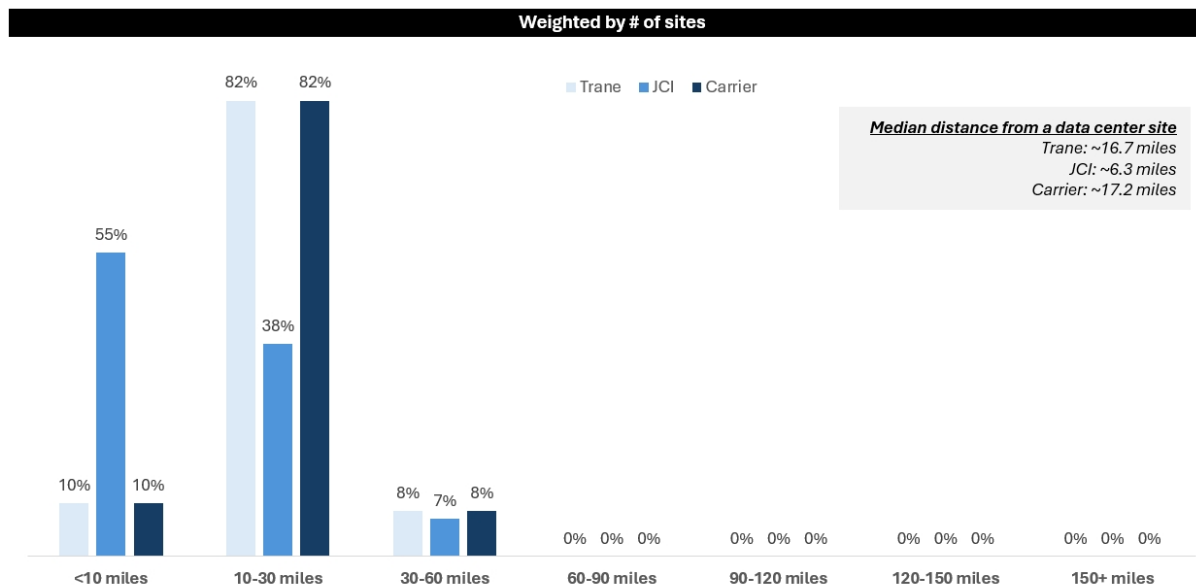
EXHIBIT 26: Iowa | Coverage breakdown weighted by capacity



Source: Aterio, Bernstein Analysis

EXHIBIT 27: Illinois | Coverage breakdown weighted by # of sites

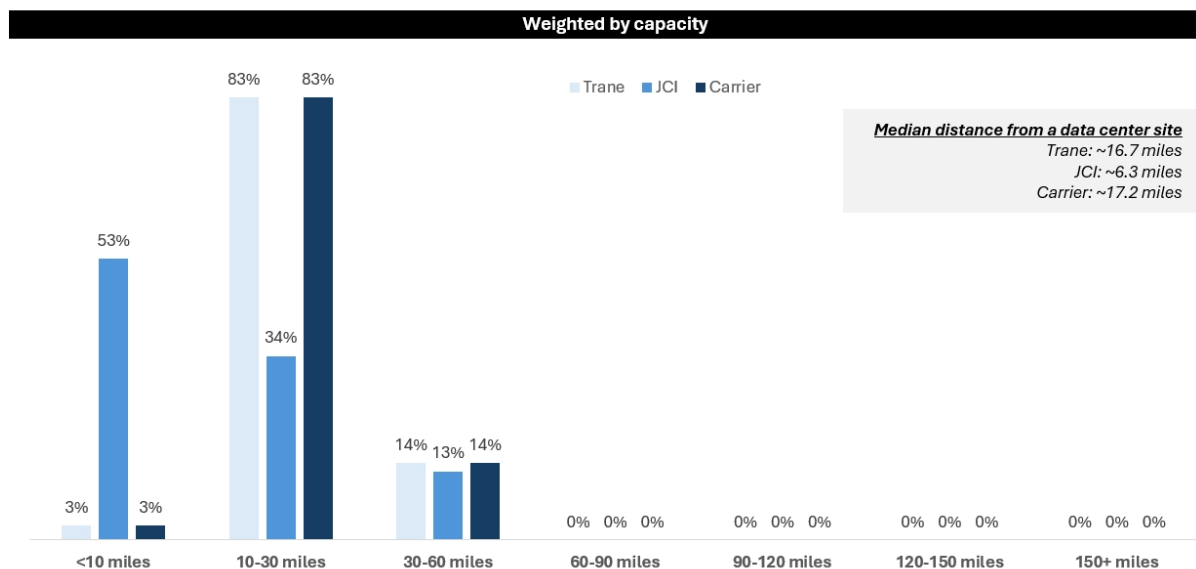
N = ~73 and ~2 GW installed capacity across Illinois



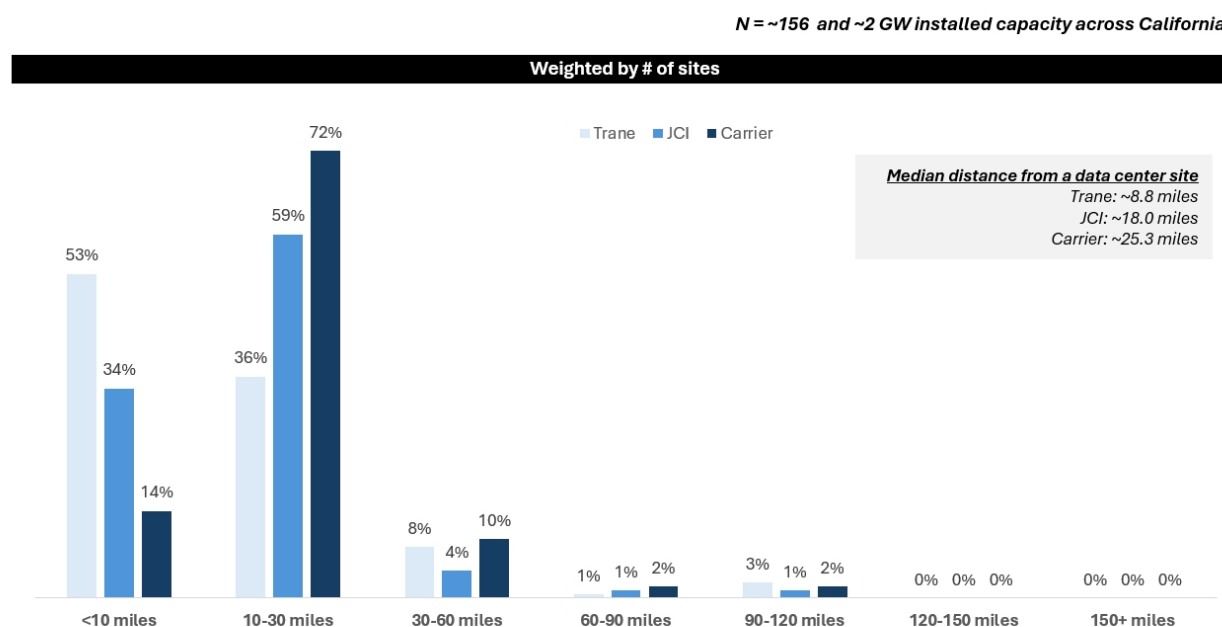
Source: Aterio, Bernstein Analysis

EXHIBIT 28: Illinois | Coverage breakdown weighted by capacity

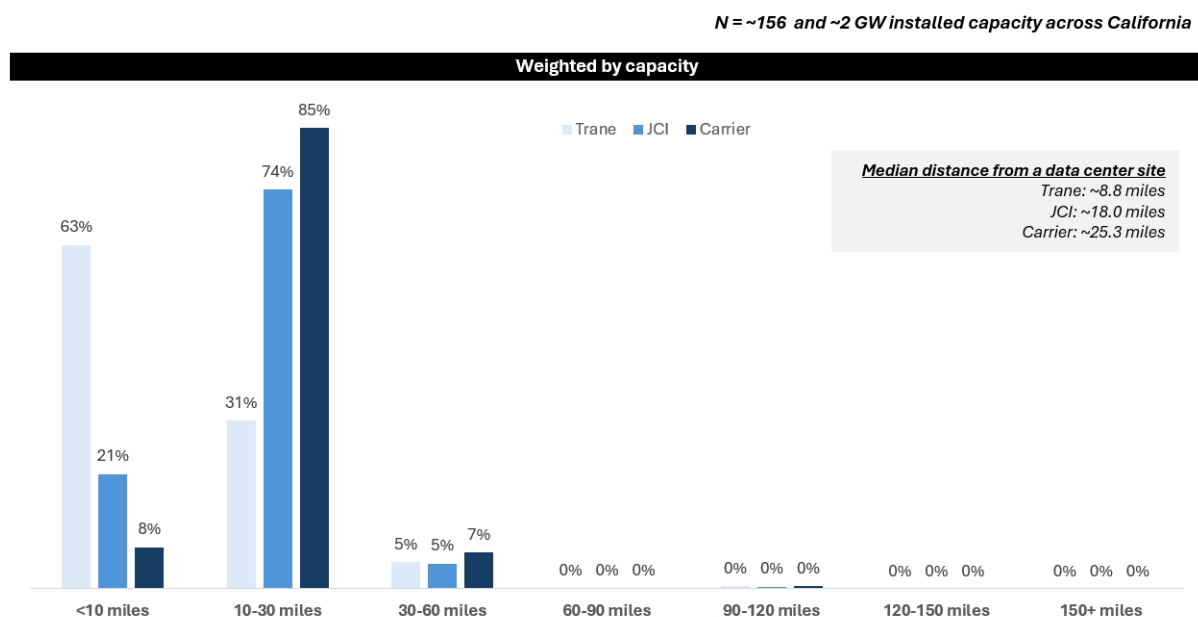
N = ~73 and ~2 GW installed capacity across Illinois



Source: Aterio, Bernstein Analysis

EXHIBIT 29: **California** | Coverage breakdown weighted by # of sites

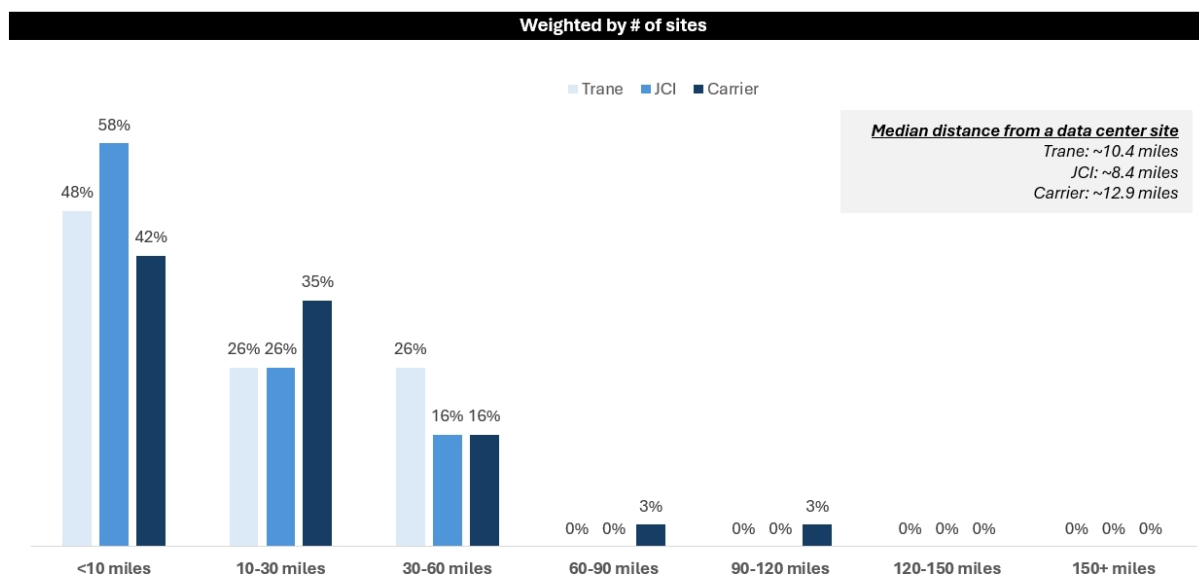
Source: Aterio, Bernstein Analysis

EXHIBIT 30: **California** | Coverage breakdown weighted by capacity

Source: Aterio, Bernstein Analysis

EXHIBIT 31: Tennessee | Coverage breakdown weighted by # of sites

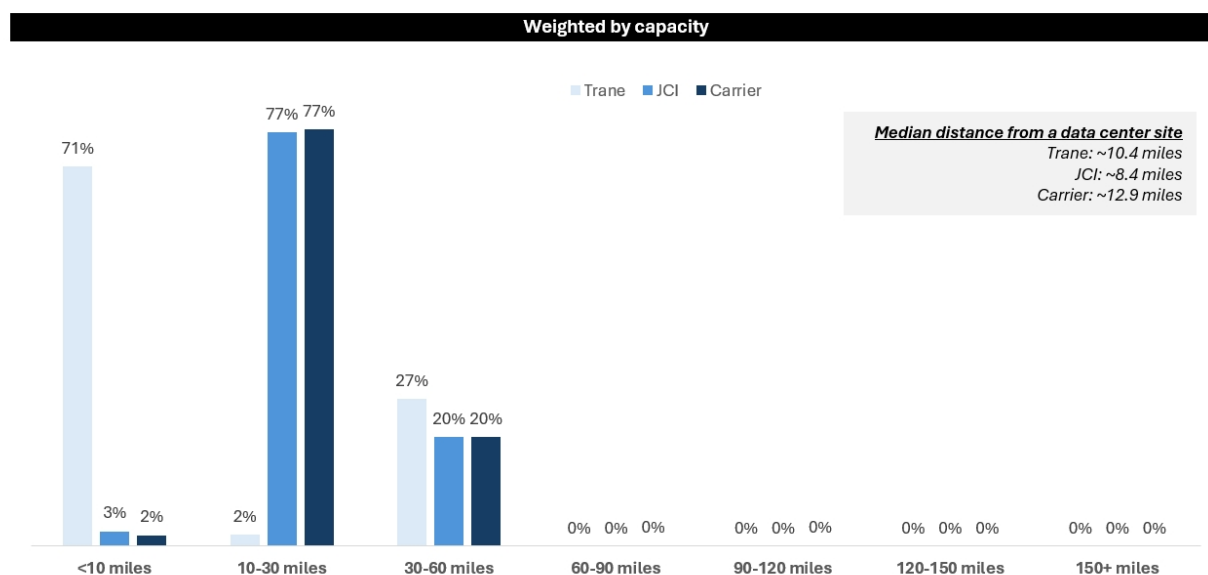
N = ~31 and ~1 GW installed capacity across Tennessee



Source: Aterio, Bernstein Analysis

EXHIBIT 32: Tennessee | Coverage breakdown weighted by capacity

N = ~31 and ~1 GW installed capacity across Tennessee



Source: Aterio, Bernstein Analysis

The authors would like to thank Sakansh Mittal for his help in preparing this report.

DISCLOSURE APPENDIX

I. REQUIRED DISCLOSURES

References to "Bernstein" or the "Firm" in these disclosures relate to the following entities: Bernstein Institutional Services LLC (April 1, 2024 onwards), Sanford C. Bernstein & Co., LLC (pre April 1, 2024), Bernstein Autonomous LLP, BSG France S.A. (April 1, 2024 onwards), Sanford C. Bernstein (Hong Kong) Limited 盛博香港有限公司, Sanford C. Bernstein (Canada) Limited, Sanford C. Bernstein (India) Private Limited (SEBI registration no. INH000006378), Sanford C. Bernstein (Singapore) Private Limited, Sanford C. Bernstein Japan KK (サンフォード・C・バーンスタイン株式会社) and analysts employed by Société Générale Africa Technologies & Services to produce Bernstein research under a Global Services Agreement in place between Bernstein and Société Générale.

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VALUATION METHODOLOGY**Trane Technologies PLC**

We value Trane at 22x NTM + 1 EBITDA of \$5.9B to reach our target price of \$575.

Johnson Controls International PLC

We value JCI using a 20x EV/EBITDA ratio and an NTM + 1 EBITDA estimate of \$5.3B to reach our target price of \$190.

Carrier Global Corporation

We value Carrier using an 18x EV/EBIT ratio and an NTM + 1 EBIT estimate of \$4.04 B to reach our target price of \$78.

RISKS**Trane Technologies PLC**

Downside: (i) Price fixing lawsuit escalates for Trane, (ii) More players achieve similar level of product innovation in CDUs / broader liquid cooling, (iii) Residential / transport numbers deteriorate further than expected, (iv) Chiller headwinds (both sales and service) outweigh liquid cooling tailwind

Johnson Controls International PLC

Downside: Lean transformation does not successfully cascade through org., Data center weakness from chiller headwinds, Operating leverage <50% based on management guidance

Carrier Global Corporation

Upside: (i) Resi / light commercial cycle turns in the US (we think this is easily the biggest driver for Carrier looking ahead), (ii) Continued growth in data centers (we'd be especially impressed if this came from CDUs) (iii) Policy changes in the UK / Germany driving heat pump uptake

Downside: (i) Escalation of price fixing lawsuit (Carrier uniquely exposed, (ii) Slowing of data center CapEx by hyperscalers, (iii) Supply of R-410A does not tighten as expected / drags out for longer

RATINGS DEFINITIONS, BENCHMARKS AND DISTRIBUTION**EQUITY RATINGS DEFINITIONS****Bernstein brand**

The Bernstein brand rates stocks based on forecasts of relative performance for the next 12 months versus the S&P 500 for stocks listed on the U.S. and Canadian exchanges, versus the Bloomberg Europe Developed Markets Large and Mid Cap Price Return Index EUR (EDME) for stocks listed on the European exchanges and emerging markets exchanges outside of the Asia

Pacific region, versus the Bloomberg Japan Large and Mid Cap Price Return Index USD (JPL) for stocks listed on the Japanese exchanges, and versus the Bloomberg Asia ex-Japan Large and Mid Cap Price Return Index (ASIA) for stocks listed on the Asian (ex-Japan) exchanges -unless otherwise specified.

The Bernstein brand has three categories of ratings:

- Outperform: Stock will outpace the market index by more than 15 pp
- Market-Perform: Stock will perform in line with the market index to within +/- 15 pp
- Underperform: Stock will trail the performance of the market index by more than 15 pp

Coverage Suspended: Coverage of a company under the Bernstein research brand has been suspended. Ratings and price targets are suspended temporarily, are no longer current, and should therefore not be relied upon.

Not Rated: A rating assigned when the stock cannot be accurately valued, or the performance of the company accurately predicted, at the present time. The covering analyst may continue to publish research reports on the company to update investors on events and developments.

Not Covered (NC) denotes companies that are not under coverage.

Bernstein brand stock ratings are based on a 12-month time horizon.

Autonomous brand – common stocks

The Autonomous brand rates common stocks as indicated below. As our benchmarks we use the Bloomberg Europe 600 Financials Price Return Index (E600BK) and Bloomberg Europe Dev Mkt Financials Large and Mid Cap Price Ret Index EUR (EDMFI) index for developed European banks and Payments, the Bloomberg Europe 600 Insurance Price Return Index (E600IN) for European insurers, the S&P 500 and S&P Financials for US banks and Payments coverage, S5LIFE for US Insurance, the S&P Insurance Select Industry (SPSIINS) for US Non-Life Insurers coverage, the Bloomberg Emerging Markets Financials Large, Mid and Small Cap Price Return Index (EMLSF) for emerging market banks and insurers and Payments, and the Bloomberg Japan Financials Large & Mid Cap Price Return Index (JPFILM) for Japanese Financials. Ratings are stated relative to the sector (not the market).

The Autonomous brand has three categories of common stock ratings:

- Outperform (OP): Stock will outpace the relevant index by more than 10 pp
- Neutral (N): Stock will perform in line with the market index to within +/- 10 pp
- Underperform (UP): Stock will trail the performance of the relevant index by more than 10 pp

Coverage Suspended: Coverage of a company under the Autonomous research brand has been suspended. Ratings and price targets are suspended temporarily, are no longer current, and should therefore not be relied upon.

Not Rated: A rating assigned when the stock cannot be accurately valued, or the performance of the company accurately predicted, at the present time. The covering analyst may continue to publish research reports on the company to update investors on events and developments.

Those denoted as 'Feature' (e.g., Feature Outperform FOP, Feature Under Outperform FUP) are our core ideas.

Not Covered (NC) denotes companies that are not under coverage.

Autonomous brand common stock ratings are based on a 12-month time horizon.

Autonomous brand – preferred stocks

The Autonomous brand has three categories of preferred stock ratings:

- Outperform (OP): The total return of the preferred instrument is expected to outperform preferred securities of other issuers operating in similar sectors or rating categories over the next six months.

- **Neutral (N):** The total return of the preferred instrument is expected to perform in line with preferred securities of other issuers operating in similar sectors or rating categories over the next six months.
- **Underperform (UP):** The total return of the preferred instrument is expected to underperform preferred securities of other issuers operating in similar sectors or rating categories over the next six months.

Autonomous preferred stock ratings are based on a 6-month time horizon.

AUTONOMOUS CREDIT RESEARCH

Where this report contains investment recommendations for credit instruments, as defined in article 3(1)(35) of the Market Abuse Regulation, the information below is presented to comply with its disclosure requirements.

The report may also include reference(s) to published opinions by other Autonomous or Bernstein analysts covering the equity securities of the issuer(s) referenced herein. Please note an investment recommendation for credit instruments published by the author(s) of this report may differ from the published view of the analyst covering equity securities for the issuer(s) contained in this report and vice versa.

CREDIT RATINGS DEFINITIONS

The Autonomous brand has three categories of credit ratings:

- **Credit Outperform (C-OP):** The total return of the Reference Credit Instrument is expected to outperform the credit spread of bonds of other issuers operating in similar sectors or rating categories over the next six months.
- **Credit Neutral (C-N):** The total return of the Reference Credit Instrument is expected to perform in line with the credit spread of bonds of other issuers operating in similar sectors or rating categories over the next six months.
- **Credit Underperform (C-UP):** The total return of the Reference Credit Instrument is expected to underperform the credit spread of bonds of other issuers operating in similar sectors or rating categories over the next six months.

Autonomous credit ratings are based on a 6-month time horizon.

A list of all investment recommendations produced by the author(s) of this report alongside credit ratings history are available upon request.

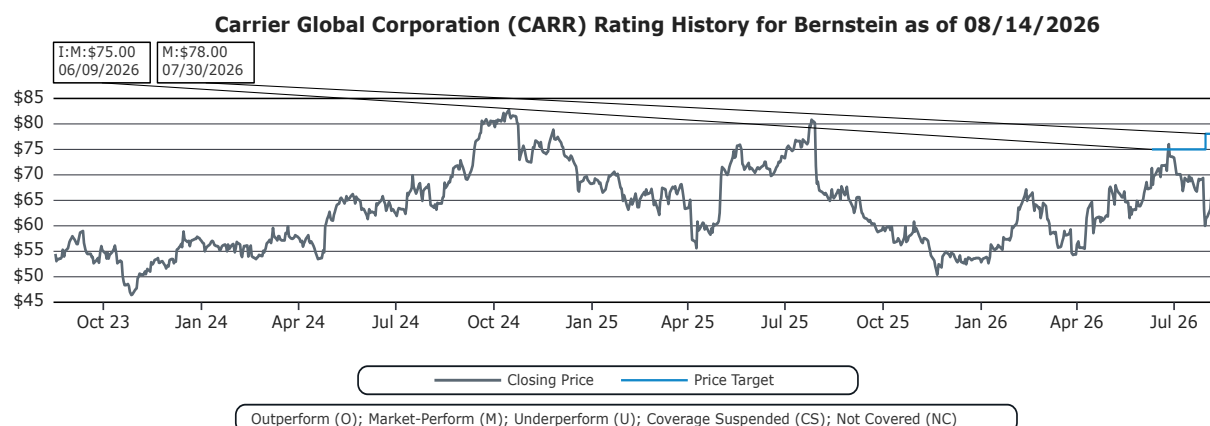
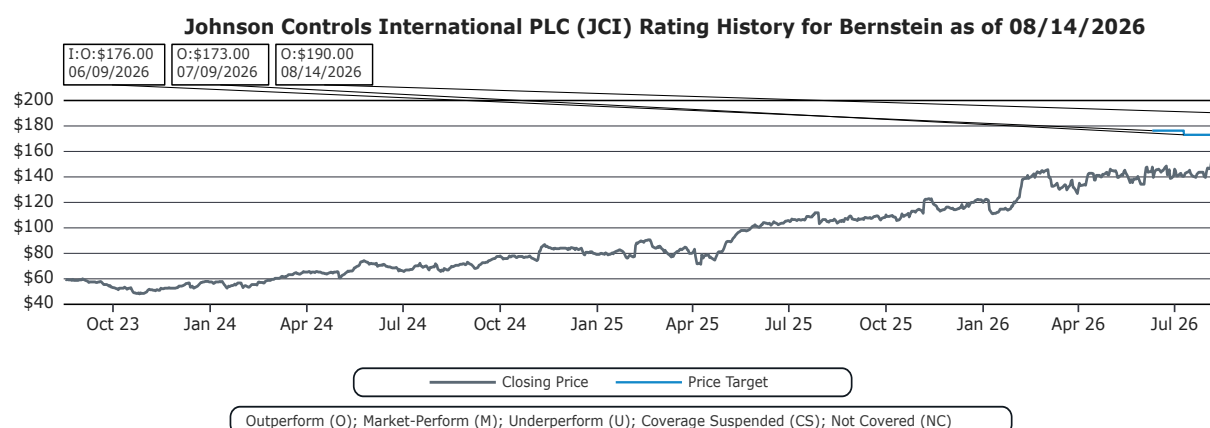
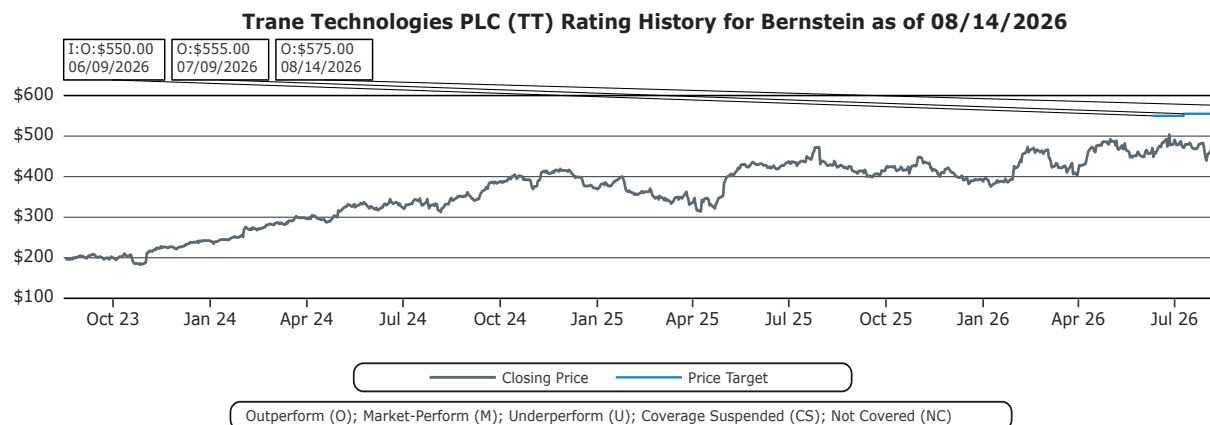
It is at the sole discretion of the Firm as to when to initiate, update and cease research coverage. The Firm has established, maintains and relies on information barriers to control the flow of information contained in one or more areas (i.e. the private side) within the Firm, and into other areas, units, groups or affiliates (i.e. public side) of the Firm

DISTRIBUTION OF EQUITY RATINGS/INVESTMENT BANKING SERVICES

Equity Rating	Market Abuse Regulation (MAR) and FINRA Rating Category	Global Rating Distribution	Investment Banking Relationships*
Outperform	BUY	51.2%	15.3%
Market-Perform (Bernstein Brand) Neutral (Autonomous Brand)	HOLD	35.8%	16.2%
Underperform	SELL	13.1%	13.6%

* These figures represent the percentage of companies within each equity rating category for which affiliates of Bernstein have provided investment banking services within the previous 12 months.

As of June 30, 2026. All figures are updated quarterly.

PRICE CHARTS / RATINGS AND PRICE TARGET HISTORY

All price target and closing price data in the chart(s) above are denominated in the currency noted in the Ticker Table of this report.

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